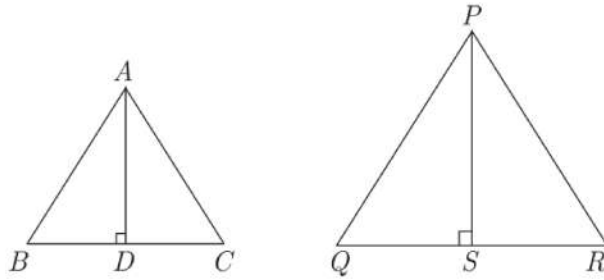


Chapter

Similarity Of Triangles

1. In the given diagram, $\triangle ABC \sim \triangle PQR$ and $\frac{AD}{PS} = \frac{3}{8}$. The value of $AB : PQ$ is:



(a) $8 : 3$

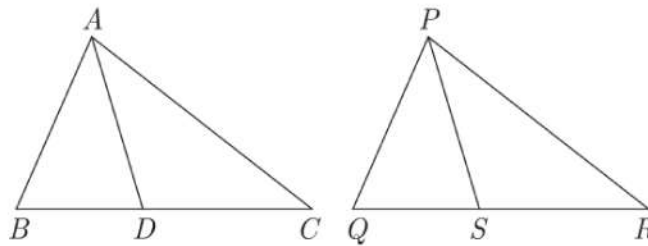
(b) $3 : 5$

(c) $3 : 8$

(d) $5 : 8$

SQP 2024

2. In the given diagram, $\triangle ABC \sim \triangle PQR$. IF AD and PS are bisectors of $\angle BAC$ and $\angle QPR$ respectively then:



(a) $\triangle ABC \sim \triangle PQS$

(b) $\triangle ABD \sim \triangle PQS$

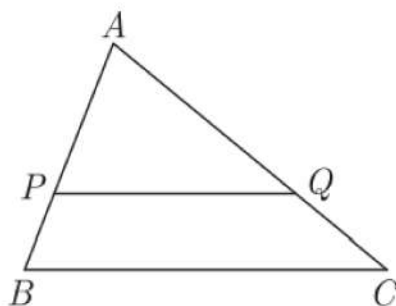
(c) $\triangle ABD \sim \triangle PSR$

(d) $\triangle ABC \sim \triangle PSR$

Main 2024

3. In the given figure $\angle BAP = \angle DCP = 70^\circ$,

$PC = 6$ cm and $CA = 4$ cm, then $PD : DB$ is :



(a) $2 : 1$

(b) $1 : \sqrt{2}$

(c) $\sqrt{2} : 1$

(d) $1 : 2$

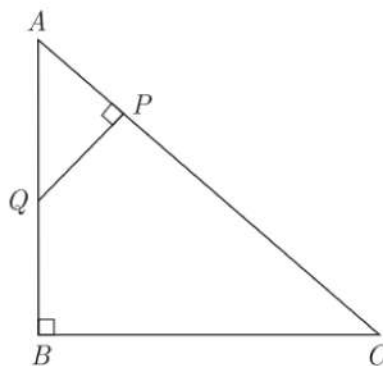
Comp 2023

6. In given triangle $\triangle ABC$ angle $\angle ABC$ is right angle and $QP \perp AC$:

(i) Show that $\triangle ABC \sim \triangle APQ$.

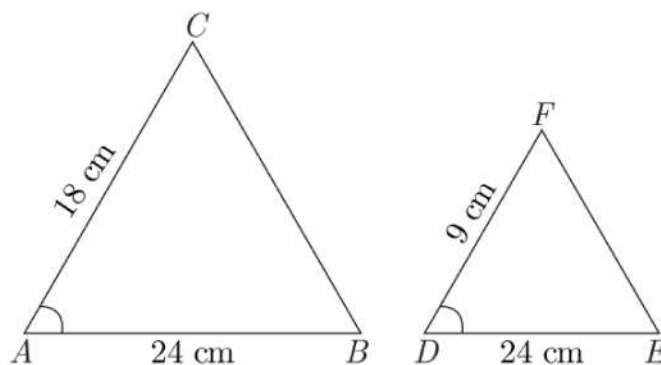
(ii) if $AP = 5$, $PQ = 12$, $BC = 36$, find the lengths of AB and AC

(iii) Find area of $\triangle APQ$: area of $\triangle ABC$.



Comp 2023

7. In the given figure $AB = 24$ cm, $AC = 18$ cm, $DE = 12$ cm, $DF = 9$ cm and $\angle BAC = \angle ADF$, Then $\triangle ABC \sim \triangle DEF$ by the condition :



(a) AAA

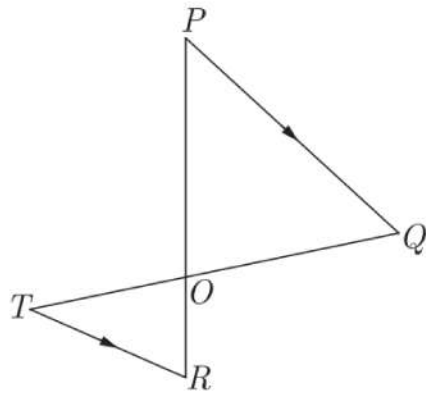
(b) SAS

(c) SSS

(d) AAS

Main 2022 Sem I

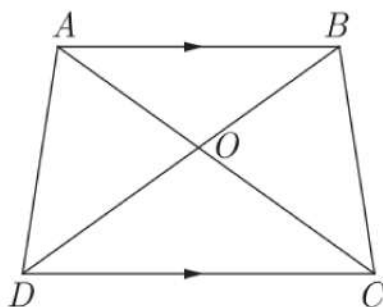
8. In the given figure PQ is parallel to TR , then by using condition of similarity:



- (a) $\frac{PQ}{RT} = \frac{OP}{OT} = \frac{OQ}{OR}$
- (b) $\frac{PQ}{RT} = \frac{OP}{OR} = \frac{OQ}{OT}$
- (c) $\frac{PQ}{RT} = \frac{OR}{OP} = \frac{OQ}{OT}$
- (d) $\frac{PQ}{RT} = \frac{OP}{OR} = \frac{OT}{OQ}$

Main 2022 Sem I

9. If $ABCD$ is a trapezium with AB parallel to DC . Then the triangle similar to $\triangle AOB$ is :



- (a) $\triangle ADB$
- (b) $\triangle ACB$
- (c) $\triangle COD$
- (d) $\triangle COB$

Main 2022 Sem I

Solution

1.

Ans: (c) 3 : 8

Explanation:

Since $\triangle ABC \sim \triangle PQR$, the ratios of the corresponding sides are equal.

$$\frac{AD}{PS} = \frac{AB}{PQ} = \frac{AC}{PR}$$

Given,

$$\frac{AD}{PS} = \frac{3}{8}$$

Therefore,

$$\frac{AB}{PQ} = \frac{3}{8}$$

$$AB : PQ = 3 : 8$$

Thus, option (c) is the correct answer.

2.

Ans: (b) $\triangle ABD \sim \triangle PQS$

Explanation:

Since $\triangle ABC \sim \triangle PQR$, corresponding angles are equal.

$$\angle A = \angle P$$

As AD and PS are angle bisectors,

$$\frac{1}{2}\angle A = \frac{1}{2}\angle P$$

$$\angle BAD = \angle QPS$$

Also, from the similarity of the triangles,

$$\angle B = \angle Q$$

Thus, in $\triangle ABD$ and $\triangle PQS$,

$$\angle BAD = \angle QPS \quad \text{and} \quad \angle B = \angle Q$$

Hence, by AA similarity criterion,

$$\triangle ABD \sim \triangle PQS$$

Thus, option (b) is the correct answer.

3.

Ans: (c) 3 : 2

Explanation:

In $\triangle PDC$ and $\triangle PBA$,

$$\angle DCP = \angle BAP = 70^\circ$$

Also, the two triangles share a common angle at P .

$$\angle DPC = \angle BPA$$

Therefore,

$$\triangle PDC \sim \triangle PBA \quad (\text{by AA similarity})$$

Hence, corresponding sides of similar triangles are proportional:

$$\frac{PC}{PA} = \frac{PD}{PB}$$

Now,

$$PA = AC + CP = 4 + 6 = 10 \text{ cm}$$

So,

$$\frac{6}{10} = \frac{PD}{PD + DB}$$

$$6(PD + DB) = 10PD$$

$$6PD + 6DB = 10PD$$

$$6DB = 4PD$$

$$\frac{PD}{DB} = \frac{6}{4} = \frac{3}{2}$$

Final Answer:

$$PD : DB = 3 : 2$$

Thus, option (c) is the correct answer.

4.

Ans:

(i) To prove $\triangle GED \sim \triangle GBF$

In $\triangle GED$ and $\triangle GBF$:

- $\angle EGD = \angle BGF$ (vertically opposite angles)
- $\angle GED = \angle GBF$ (alternate interior angles, since $DE \parallel BF$)

Hence, by **AA similarity criterion**,

$$\triangle GED \sim \triangle GBF$$

(ii) To find DE

Since corresponding sides of similar triangles are proportional,

$$\frac{GE}{GB} = \frac{DE}{BF}$$

Substituting the given values,

$$\frac{8}{16} = \frac{DE}{30}$$

$$DE = \frac{8}{16} \times 30 = 15 \text{ cm}$$

(iii) To find $DB : AB$

In $\triangle ABC$, since $DE \parallel AC$, by the **Basic Proportionality Theorem**,

$$\frac{DE}{AC} = \frac{DB}{AB}$$

Substituting values,

$$\frac{15}{24} = \frac{DB}{AB}$$

$$DB : AB = 5 : 8$$

Final Answers:

- $\triangle GED \sim \triangle GBF$
- $DE = 15$ cm
- $DB : AB = 5 : 8$

5.

Ans: (b) $1 : \sqrt{2}$

Explanation:

Since $PQ \parallel BC$,

$$\triangle ABC \sim \triangle APQ$$

For similar triangles, the ratio of their areas is equal to the square of the ratio of corresponding sides:

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle APQ} = \left(\frac{AB}{AP}\right)^2$$

Given that the triangle is divided into two equal areas,

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle APQ} = 2$$

So,

$$2 = \left(\frac{AB}{AP} \right)^2$$

$$\sqrt{2} = \frac{AB}{AP}$$

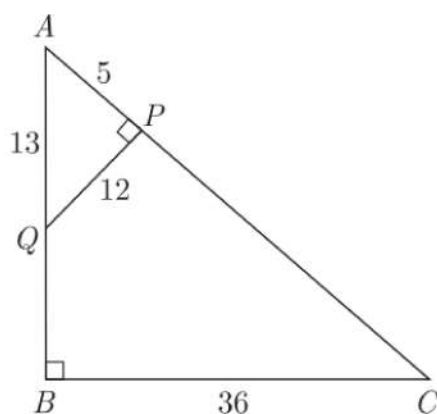
$$AP : AB = 1 : \sqrt{2}$$

Thus, option (b) is the correct answer.

6.

Ans:

We have redrawn the given figure as below.



(i) $\triangle ABC \sim \triangle APQ$

In $\triangle APQ$ and $\triangle ABC$,

Due to common angle,

$$\angle A = \angle A$$

Due to right angle,

$$\angle APQ = \angle ABC$$

By AA-axiom similarity we have

$$\triangle APQ \sim \triangle ABC$$

(ii) Length of AB and AC

$$AQ = \sqrt{5^2 + 12^2}$$

$$= 13 \text{ cm}$$

If two triangles are similar, then their corresponding sides are proportional.

$$\frac{AP}{AB} = \frac{AQ}{AC} = \frac{PQ}{BC}$$

Substituting $AP = 5, PQ = 12, BC = 36$ and $AQ = 13$ cm we have

$$\frac{5}{AB} = \frac{12}{36} = \frac{AQ}{AC}$$

$$AB = \frac{5 \times 36}{12} = 15 \text{ cm}$$

In right $\triangle ABC$,

$$AC^2 = AB^2 + BC^2$$

$$= 15^2 + 36^2$$

$$= 225 + 1296$$

$$= 1521$$

$$AC = \sqrt{1521}$$

$$= 39 \text{ units}$$

Since areas of two similar triangles are proportional to the square of their corresponding sides. Thus

$$\frac{\text{area of } \triangle APQ}{\text{area of } \triangle ABC} = \frac{AP^2}{AB^2}$$

$$= \frac{5^2}{15^2}$$

$$= \frac{25}{225} = \frac{1}{9}$$

Area of $\triangle APQ$: area of $\triangle ABC = 1 : 9$

$$\angle C = \angle A$$

By AA-axiom similarity we have

$$\triangle FCD \sim \triangle FAB$$

If two triangles are similar, then their corresponding sides are proportional.

Thus

$$\begin{aligned}\frac{FC}{FA} &= \frac{CD}{AB} = \frac{FD}{FB} \\ \frac{6}{6+4} &= \frac{y}{5} \\ CD &= y \\ &= \frac{6}{10} \times 5 = 3 \text{ cm}\end{aligned}$$

(iii) Area $\triangle CDF$: area of $\triangle ABF$.

Since areas of two similar triangles are proportional to the square of their corresponding sides. Thus

$$\begin{aligned}\frac{\text{Area}(\triangle CDF)}{\text{Area}(\triangle ABF)} &= \frac{CD^2}{AB^2} \\ &= \frac{3^2}{5^2} = \frac{9}{25}\end{aligned}$$

7.

Ans: (b) SAS

Explanation:

We know that if **one angle of a triangle is equal to one angle of another triangle** and the **sides including those angles are proportional**, then the two triangles are similar by the **SAS similarity criterion**.

Here,

$$\angle BAC = \angle EDF$$

Also,

$$\frac{AB}{AC} = \frac{24}{18} = \frac{4}{3} \quad \text{and} \quad \frac{DF}{DE} = \frac{9}{12} = \frac{3}{4}$$

Thus, the sides including the equal angles are proportional.

Therefore,

$\triangle ABC \sim \triangle DEF$ by SAS similarity criterion

Hence, option (b) is the correct answer.

8.

Ans: (b)

$$\frac{PQ}{RT} = \frac{OP}{OR} = \frac{OQ}{OT}$$

Explanation:

Since $PQ \parallel RT$

- $\angle T = \angle Q$ (alternate interior angles)
- $\angle R = \angle P$ (vertically opposite angles)

Hence, by **AA similarity criterion**,

$$\triangle PQO \sim \triangle RTO$$

Therefore, the corresponding sides of similar triangles are proportional:

$$\frac{PQ}{RT} = \frac{OP}{OR} = \frac{OQ}{OT}$$

Thus, option (b) is the correct answer.

9.

Ans: (c) $\triangle COD$

Explanation:

Since $AB \parallel DC$,

- $\angle ABO = \angle CDO$ (alternate interior angles)
- $\angle BAO = \angle DCO$ (alternate interior angles)

Also,

- $\angle AOB = \angle COD$ (vertically opposite angles)

Thus, two corresponding angles of $\triangle AOB$ and $\triangle COD$ are equal.

$$\therefore \triangle AOB \sim \triangle COD \quad (\text{by AA similarity criterion})$$

Hence, option (c) is the correct answer.